

# Supplementary Information

## Spectral Diagnostic Trajectories: A Hypergraph Framework for Dynamic Clinical Decision Support and Operational Optimization

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### Contents

|          |                                                                                                 |          |
|----------|-------------------------------------------------------------------------------------------------|----------|
| <b>1</b> | <b>Supplementary Note 1: Mathematical Proof of Theorem 1 (Algebraic Connectivity Guarantee)</b> | <b>2</b> |
| 1.1      | Statement of Theorem 1 . . . . .                                                                | 2        |
| 1.2      | Proof of Part (i) . . . . .                                                                     | 2        |
| 1.3      | Proof of Part (ii) . . . . .                                                                    | 3        |
| 1.4      | Proof of Part (iii) . . . . .                                                                   | 3        |
| <b>2</b> | <b>Supplementary Note 2: Mathematical Proof of Theorem 2 (Cheeger Contamination Bounds)</b>     | <b>4</b> |
| 2.1      | Statement of Theorem 2 . . . . .                                                                | 4        |
| 2.2      | Proof . . . . .                                                                                 | 4        |
| <b>3</b> | <b>Supplementary Note 3: Algorithm Pseudocode &amp; Complexity Analysis</b>                     | <b>5</b> |
| 3.1      | Computational Complexity Comparison . . . . .                                                   | 5        |
| <b>4</b> | <b>Supplementary Note 4: Detailed Baseline Implementations</b>                                  | <b>6</b> |
| <b>5</b> | <b>Supplementary Note 5: Vignette Construction &amp; Simulation Parameters</b>                  | <b>7</b> |
| <b>6</b> | <b>Supplementary Note 6: Comprehensive Hyperparameter Specifications</b>                        | <b>8</b> |

# 1 Supplementary Note 1: Mathematical Proof of Theorem 1 (Algebraic Connectivity Guarantee)

## 1.1 Statement of Theorem 1

Let  $G_t = (V_t, E_t, W_t)$  be the active clinical diagnostic hypergraph with vertex degree matrix  $D_v$ , hyperedge degree matrix  $D_e$ , hyperedge weight matrix  $W_t$ , and normalized hypergraph Laplacian  $L_t = I - D_v^{-1/2} H_t W_t D_e^{-1} H_t^\top D_v^{-1/2}$ .

- (i) **Connectivity Guarantee:**  $\lambda_2(L_t) > 0$  if and only if  $G_t$  is connected, corresponding to a single, structurally unified diagnostic hypothesis.
- (ii) **Monotonic Evidence Accumulation:** If a new evidence node  $v_{\text{new}}$  is connected to existing active nodes  $u \in V_t$  with edge weights  $w \geq w_{\min} > 0$ , then:

$$\lambda_2(L_{t+1}) \geq \lambda_2(L_t) - \mathcal{O}\left(\frac{1}{|V_t|}\right), \quad (1)$$

ensuring spectral stability during corroborating evidence gathering.

- (iii) **Hypothesis Bifurcation (Pivot):** If the clinician introduces a set of queries and evidence  $V_{\text{new}}$  representing an alternative differential diagnosis such that inter-cluster hyperedge weight between  $V_{\text{old}}$  and  $V_{\text{new}}$  satisfies  $\sum_{e \in E_{\text{cut}}} w(e) \leq \epsilon$ , then:

$$\lambda_2(L_t) \leq \frac{2\epsilon}{\min(\text{vol}(V_{\text{old}}), \text{vol}(V_{\text{new}}))}. \quad (2)$$

## 1.2 Proof of Part (i)

By the quadratic form of the normalized hypergraph Laplacian (Chung, 1997; Zhou et al., 2006):

$$f^\top L_t f = \frac{1}{2} \sum_{e \in E_t} \frac{w(e)}{\delta(e)} \sum_{u,v \in e} \left( \frac{f(u)}{\sqrt{d(u)}} - \frac{f(v)}{\sqrt{d(v)}} \right)^2. \quad (3)$$

Because  $w(e) > 0$  and  $\delta(e) = |e| \geq 2$ ,  $f^\top L_t f \geq 0$  for all  $f \in \mathbb{R}^{|V_t|}$ . Equality  $f^\top L_t f = 0$  holds if and only if for every hyperedge  $e \in E_t$  and every pair of vertices  $u, v \in e$ :

$$\frac{f(u)}{\sqrt{d(u)}} = \frac{f(v)}{\sqrt{d(v)}}. \quad (4)$$

If  $G_t$  is connected, any two vertices  $x, y \in V_t$  are connected through a sequence of intersecting hyperedges. By induction,  $f(x)/\sqrt{d(x)} = f(y)/\sqrt{d(y)}$  across the entire graph. Thus, the harmonic eigenspace associated with eigenvalue 0 has dimension exactly 1, spanned by the unit vector  $\phi_1 = D_v^{1/2} \mathbf{1} / \|D_v^{1/2} \mathbf{1}\|_2$ .

Because the eigenvalues of  $L_t$  are non-negative and sorted in ascending order  $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_{|V_t|}$ , the multiplicity of eigenvalue 0 equals the number of connected components  $k$ . Therefore,  $\lambda_2(L_t) > 0$  if and only if  $k = 1$  (the hypergraph consists of a single connected component).  $\square$

### 1.3 Proof of Part (ii)

By the Courant-Fischer min-max theorem:

$$\lambda_2(L_t) = \min_{f \perp D_v^{1/2} \mathbf{1}, f \neq 0} \frac{f^\top L_t f}{f^\top f}. \quad (5)$$

Let  $V_{t+1} = V_t \cup \{v_{\text{new}}\}$ . When a corroborating node is added with weights bounded below by  $w_{\min}$ , the quadratic form in Eq. (3) receives non-negative additive terms corresponding to the new hyperedges:

$$f^\top L_{t+1} f \geq f^\top L_t f - \delta_{\text{pert}}(f). \quad (6)$$

By Weyl's eigenvalue perturbation theorem, the perturbation to  $\lambda_2$  induced by adding a node to a graph of size  $|V_t|$  is bounded by the maximum normalized degree alteration, which scales as  $\mathcal{O}(1/|V_t|)$ . Hence,  $\lambda_2(L_{t+1}) \geq \lambda_2(L_t) - \mathcal{O}(1/|V_t|)$ .  $\square$

### 1.4 Proof of Part (iii)

Let the active vertex set be partitioned into  $V_t = V_{\text{old}} \cup V_{\text{new}}$  with  $V_{\text{old}} \cap V_{\text{new}} = \emptyset$ . Define the volume of a subset  $S \subseteq V_t$  as  $\text{vol}(S) = \sum_{v \in S} d(v)$ . Construct a test vector  $g \in \mathbb{R}^{|V_t|}$  such that:

$$g(u) = \begin{cases} \frac{1}{\text{vol}(V_{\text{old}})}, & \text{if } u \in V_{\text{old}}, \\ -\frac{1}{\text{vol}(V_{\text{new}})}, & \text{if } u \in V_{\text{new}}. \end{cases} \quad (7)$$

Let  $f = D_v^{1/2} g$ . Then:

$$f^\top D_v^{1/2} \mathbf{1} = g^\top D_v \mathbf{1} = \sum_{u \in V_{\text{old}}} \frac{d(u)}{\text{vol}(V_{\text{old}})} - \sum_{v \in V_{\text{new}}} \frac{d(v)}{\text{vol}(V_{\text{new}})} = 1 - 1 = 0, \quad (8)$$

so  $f$  is strictly orthogonal to the harmonic ground state  $\phi_1$ . By Courant-Fischer (Eq. (5)):

$$\lambda_2(L_t) \leq \frac{f^\top L_t f}{f^\top f} = \frac{g^\top \mathcal{L}_t g}{g^\top D_v g}, \quad (9)$$

where  $\mathcal{L}_t = D_v - H_t W_t D_e^{-1} H_t^\top$ . The denominator evaluates to:

$$g^\top D_v g = \frac{1}{\text{vol}(V_{\text{old}})} + \frac{1}{\text{vol}(V_{\text{new}})} \geq \frac{1}{\min(\text{vol}(V_{\text{old}}), \text{vol}(V_{\text{new}}))}. \quad (10)$$

The numerator evaluates across boundary cut hyperedges:

$$g^\top \mathcal{L}_t g = \frac{1}{2} \sum_{e \in E_{\text{cut}}} \frac{w(e)}{\delta(e)} \sum_{u, v \in e} (g(u) - g(v))^2 \leq \sum_{e \in E_{\text{cut}}} w(e) \cdot \left( \frac{1}{\text{vol}(V_{\text{old}})} + \frac{1}{\text{vol}(V_{\text{new}})} \right)^2. \quad (11)$$

Substituting  $\sum_{e \in E_{\text{cut}}} w(e) \leq \epsilon$  yields:

$$\lambda_2(L_t) \leq \frac{2\epsilon}{\min(\text{vol}(V_{\text{old}}), \text{vol}(V_{\text{new}}))}. \quad (12)$$

Thus, as the alternative hypothesis forms a distinct, weakly connected cluster ( $\epsilon \rightarrow 0$ ), algebraic connectivity collapses toward zero.  $\square$

## 2 Supplementary Note 2: Mathematical Proof of Theorem 2 (Cheeger Contamination Bounds)

### 2.1 Statement of Theorem 2

Let  $\tau_{\text{pivot}} > 0$  be the pivot sensitivity threshold. When the Spectral Interference Gate fires ( $\Delta\lambda_2 > \tau_{\text{pivot}}$ ), the Fiedler vector partition  $V_t = V_{\text{active}} \cup V_{\text{stale}}$  with severed boundary hyperedges satisfies:

$$h(G_t) \leq \sqrt{2\lambda_2(L_t)} \leq \sqrt{2\tau_{\text{pivot}}}, \quad (13)$$

and the residual transition probability mass from  $V_{\text{stale}}$  into prospective retrieval candidates is identically bounded by  $\tau_{\text{pivot}}$ .

### 2.2 Proof

The Cheeger constant (conductance) of a hypergraph partition is defined as:

$$h(G_t) = \min_{\emptyset \subsetneq S \subset V_t} \frac{\text{vol}(\partial S)}{\min(\text{vol}(S), \text{vol}(V_t \setminus S))}, \quad (14)$$

where  $\text{vol}(\partial S) = \sum_{e \in E_t, e \cap S \neq \emptyset, e \cap (V_t \setminus S) \neq \emptyset} w(e)$ . By the discrete Cheeger inequality for normalized hypergraph Laplacians (Chung, 1997):

$$2h(G_t) \geq \lambda_2(L_t) \geq \frac{h^2(G_t)}{2} \implies h(G_t) \leq \sqrt{2\lambda_2(L_t)}. \quad (15)$$

Because the normalized cut capacity is bounded by  $2h(S)$ , severing the boundary hyperedges completely eliminates all direct transition paths between  $V_{\text{stale}}$  and  $V_{\text{active}}$ .

During subsequent random walk diffusion on the modified graph, the transition probability matrix satisfies  $P = D_v^{-1}A$ . Because all edges spanning  $V_{\text{stale}}$  and  $V_{\text{active}}$  have been severed, the transition matrix decomposes into a block diagonal structure:

$$P_{\text{post-cut}} = \begin{bmatrix} P_{\text{stale}} & 0 \\ 0 & P_{\text{active}} \end{bmatrix}. \quad (16)$$

Consequently, the diffusion transition probability mass from any node in  $V_{\text{stale}}$  into  $V_{\text{active}}$  is identically zero:

$$P(v_{\text{active}} \mid v_{\text{stale}}) = 0. \quad (17)$$

When the Spectral Interference Gate fires ( $\Delta\lambda_2(t) = \lambda_2(L_{t-1}) - \lambda_2(L_t) > \tau_{\text{pivot}}$ ), the post-pivot eigenvalue satisfies  $\lambda_2(L_t) \leq \tau_{\text{pivot}}$ . The maximum residual spectral contamination leakage  $\mathcal{C}_{\text{leak}}$  in candidate retrieval scores is bounded by the Rayleigh quotient across the severed boundary:

$$\mathcal{C}_{\text{leak}} \leq \lambda_2(L_t) \leq \tau_{\text{pivot}}. \quad (18)$$

Thus, information leakage is strictly bounded by  $\tau_{\text{pivot}}$ .  $\square$

### 3 Supplementary Note 3: Algorithm Pseudocode & Complexity Analysis

**Algorithm:** *Spectral Diagnostic Trajectory (SDT) Engine*

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**Input:** Longitudinal patient record stream  $\mathcal{S}$ , current query  $q_t$ , prior hypergraph  $G_{t-1} = (V_{t-1}, E_{t-1})$ , threshold  $\tau_{\text{pivot}}$ , knapsack budget  $C_{\max}(t)$ .

**Output:** Filtered candidate document set  $\mathcal{D}_t^*$ , updated hypergraph  $G_t$ .

- 1: Extract new clinical entities  $V_{\text{new}}$  from query  $q_t$  and streaming telemetry (via Mamba SSM).
- 2: Construct updated incidence matrix  $H_t$  by binding  $V_{\text{new}}$  to active hyperedges  $E_t$ .
- 3: Form normalized hypergraph Laplacian  $L_t = I - D_v^{-1/2} H_t W_t D_e^{-1} H_t^\top D_v^{-1/2}$ .
- 4: Compute algebraic connectivity  $\lambda_2(L_t)$  and Fiedler vector  $v_2$  via restarted Lanczos on  $\mathcal{P}_t = I - L_t$ .
- 5: Calculate Spectral Pivot Metric  $\Delta\lambda_2(t) = \lambda_2(L_{t-1}) - \lambda_2(L_t)$ .
- 6: **if**  $\Delta\lambda_2(t) > \tau_{\text{pivot}}$  **then** // Diagnostic pivot detected
- 7:   Partition  $V_t = V_{\text{active}} \cup V_{\text{stale}}$  where  $V_{\text{active}} = \{u \mid v_2(u) \geq 0\}$ .
- 8:   Sever all cut hyperedges:  $E_t \leftarrow E_t \setminus \{e \in E_t \mid e \cap V_{\text{active}} \neq \emptyset \wedge e \cap V_{\text{stale}} \neq \emptyset\}$ .
- 9:   Attenuate  $V_{\text{stale}}$  in working memory cache.
- 10: **else**
- 11:    $V_{\text{active}} \leftarrow V_t$ .
- 12: **end if**
- 13: Execute spectrally-biased random walk from  $V_{\text{active}}$  on Voronoi-partitioned corpus index to retrieve candidates  $\mathcal{C}_t$ .
- 14: Solve Sweller knapsack optimization over candidates  $\mathcal{C}_t$ :

$$\mathcal{D}_t^* = \arg \max_{\mathcal{D} \subseteq \mathcal{C}_t} \sum_{d \in \mathcal{D}} R(d) \quad \text{s.t.} \quad \sum_{d \in \mathcal{D}} c(d) \leq C_{\max}(t).$$

15: Present  $\mathcal{D}_t^*$  ( $|\mathcal{D}_t^*| \leq 5$ ) to clinician interface.

16: **return**  $\mathcal{D}_t^*, G_t$ .

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#### 3.1 Computational Complexity Comparison

| Operation                 | Riemannian GDT                    | Dense RAG (CMA)                   | SDT (Ours)                                            |
|---------------------------|-----------------------------------|-----------------------------------|-------------------------------------------------------|
| State Maintenance         | $O(k \cdot d^2)$ (Covariance)     | $O(k \cdot d)$ (Sliding window)   | $O( E_t  \cdot  e )$ (Incidence update)               |
| Shift / Pivot Detection   | $O(d^3)$ (Matrix Logarithm)       | None (Stateless)                  | $O( E_t )$ (Lanczos on $\mathcal{P}_t$ )              |
| Context Decoupling        | Heuristic decay $e^{-\gamma t}$   | Uniform window decay              | Cheeger bisection ( $O( V_t )$ )                      |
| Prospective Prefetch      | $O(d^2)$ (Neural JEPA)            | None                              | Biased walk ( $O(N_{\text{walk}} \cdot \text{deg})$ ) |
| Delivery Filtering        | None                              | None                              | Greedy knapsack ( $O(K_c \log K_c)$ )                 |
| <b>Total Turn Latency</b> | <b><math>\sim 128.2</math> ms</b> | <b><math>\sim 146.5</math> ms</b> | <b><math>\sim 112.9</math> ms</b>                     |

## 4 Supplementary Note 4: Detailed Baseline Implementations

**1. Control Baseline (TF-IDF).** Standard clinical keyword search utilizing term frequency-inverse document frequency (TF-IDF) feature weighting with sublinear term frequency scaling and cosine similarity ranking.

**2. BM25 Lexical Retriever.** Best Matching 25 implementation with standard parameters  $k_1 = 1.5$  and  $b = 0.75$ , modeling length-normalized document saturation over the 546,028-note corpus.

**3. CMA (Dense RAG).** Context-Memory Augmented dense retriever utilizing a PubMedBERT bi-encoder. Context history is maintained via a sliding window of the last 3 turns, decaying historical tokens uniformly.

**4. MedCPT Contrastive Retriever.** Bi-encoder architecture pretrained on PubMed search logs and clinical notes via contrastive learning. Query and document representations are mapped into a 256-dimensional dense embedding space, with retrieval driven by inner product similarity.

**5. ColBERT-Clinical Late-Interaction Retriever.** Token-level multi-vector late-interaction retriever. Evaluates maximum similarity (MaxSim) across all query token vectors  $Q$  and candidate document token vectors  $D$ :

$$S_{\text{MaxSim}}(Q, D) = \sum_{q \in Q} \max_{d \in D} (E_q \cdot E_d). \quad (19)$$

**6. GraphCare Knowledge-Graph Retriever.** Constructs a clinical document similarity graph using 5-nearest neighbors over latent representations. A 2-layer Graph Convolutional Network (GCN) executes message passing across the graph to enrich document representations with neighborhood context prior to cosine ranking.

**7. LLM-RAG / EHR-RAG Re-Ranker.** A two-stage retrieval pipeline. The first stage retrieves the top-50 candidates using lexical search. The second stage applies a learned cross-attentive MLP re-ranking model taking concatenated query-document representations to predict clinical relevance.

**8. Geodesic Diagnostic Trajectories (GDT).** Tracks clinical intent on the Riemannian manifold of  $136 \times 136$  Symmetric Positive Definite (SPD) covariance matrices. Geodesic shifts are computed via the affine-invariant metric  $\delta_R(S_1, S_2) = \|\log(S_1^{-1/2} S_2 S_1^{-1/2})\|_F$ . Context attenuation is applied through continuous exponential decay  $e^{-\gamma \Delta t}$ . Prefetching is driven by a trained Joint-Embedding Predictive Architecture (JEPA).

## 5 Supplementary Note 5: Vignette Construction & Simulation Parameters

**Clinical Vignette Distribution.** The 1,000 standardized diagnostic vignettes span four acute clinical hospital disciplines:

- **Critical Care Medicine** ( $n = 264$ ): Sepsis with septic shock, acute respiratory distress syndrome (ARDS), severe diabetic ketoacidosis with cerebral edema, cardiogenic shock, and massive gastrointestinal hemorrhage.
- **Emergency Medicine** ( $n = 250$ ): Atypical acute coronary syndromes, pulmonary embolism, aortic dissection, acute ischemic stroke, ectopic pregnancy, and undifferentiated abdominal pain.
- **Inpatient Cardiology** ( $n = 244$ ): Acute decompensated heart failure with preserved ejection fraction, acute myocarditis, cardiac tamponade, ventricular tachycardia storm, and infective endocarditis.
- **Acute Neurology** ( $n = 242$ ): Status epilepticus, acute subarachnoid hemorrhage, bacterial vs. viral meningitis, Guillain-Barré syndrome, and acute basilar artery occlusion.

**Human-Factors Calibration.** Simulated clinician navigation parameters were calibrated to empirical time-and-motion studies:

| Parameter                        | Calibrated Value                           | Literature Source                         |
|----------------------------------|--------------------------------------------|-------------------------------------------|
| Relevant Note Reading Time       | Uniform(15.0, 45.0) seconds                | Sinsky et al. (2016); Zheng et al. (2021) |
| Non-Relevant Note Screening Time | Uniform(5.0, 12.0) seconds                 | Harry et al. (2018)                       |
| Query Formulation Latency        | Uniform(8.0, 20.0) seconds                 | Zheng et al. (2021)                       |
| Maximum Queries per Session      | 10 queries                                 | Established clinical benchmark            |
| Cognitive Load Intrinsic Cost    | $c(d) = 1.0 + 0.05 \cdot \text{len}(d)$    | Sweller Cognitive Load Theory (2019)      |
| Cognitive Load Extraneous Cost   | 1.5 $\times$ for irrelevant search results | Sweller (1988); Harry et al. (2018)       |

## 6 Supplementary Note 6: Comprehensive Hyperparameter Specifications

| Component                         | Hyperparameter                                        | Value / Specification                                                               |
|-----------------------------------|-------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Mamba SSM Waveform Encoder</b> | Input telemetry sampling frequency                    | 125–250 Hz                                                                          |
|                                   | Model hidden state dimension ( $D$ )                  | 128                                                                                 |
|                                   | SSM expansion factor                                  | 2                                                                                   |
|                                   | Selective state dimension ( $N$ )                     | 16                                                                                  |
|                                   | Discretization kernel                                 | Zero-Order Hold (ZOH)                                                               |
| <b>Hypergraph Construction</b>    | Maximum active vertex count ( $ V_{\max} $ )          | 30 nodes                                                                            |
|                                   | Vertex modality partition                             | Text ( $V^{\text{text}}$ ), Codes ( $V^{\text{code}}$ ), Wave ( $V^{\text{wave}}$ ) |
|                                   | Hyperedge weight initialization                       | $W_t(e, e) = 1.0$ (co-occurrence frequency)                                         |
|                                   | Working memory decay rate                             | Spectral cut bisection (zero cross-cut weight)                                      |
| <b>Spectral Eigensolver</b>       | Target matrix                                         | Transition operator $\mathcal{P}_t = I - L_t$                                       |
|                                   | Eigensolver algorithm                                 | Implicitly Restarted Lanczos (ARPACK)                                               |
|                                   | Selection mode                                        | ‘which=‘LA’‘ (Largest Algebraic)                                                    |
|                                   | Convergence tolerance                                 | $10^{-6}$                                                                           |
|                                   | Maximum Lanczos basis size                            | 15                                                                                  |
| <b>Spectral Interference Gate</b> | Pivot sensitivity threshold ( $\tau_{\text{pivot}}$ ) | 0.35 (calibrated on validation split)                                               |
|                                   | Decoupling cut criterion                              | Sign bisection: $v_2(u) \geq 0$ vs. $v_2(u) < 0$                                    |
|                                   | Post-cut cross-boundary leakage                       | $\mathcal{C}_{\text{leak}} \leq 0.35$ (bounded by Theorem 2)                        |
| <b>Prefetching &amp; Knapsack</b> | Corpus Voronoi partitions                             | $K = 2,048$ clusters                                                                |
|                                   | Random walk length ( $N_{\text{walk}}$ )              | 20 steps                                                                            |
|                                   | Spectral bias parameter ( $\beta$ )                   | 1.5                                                                                 |
|                                   | Knapsack working memory capacity ( $C_{\max}$ )       | 5.0 normalized cognitive units                                                      |
|                                   | Maximum delivered candidates ( $k$ )                  | 5 documents per turn                                                                |